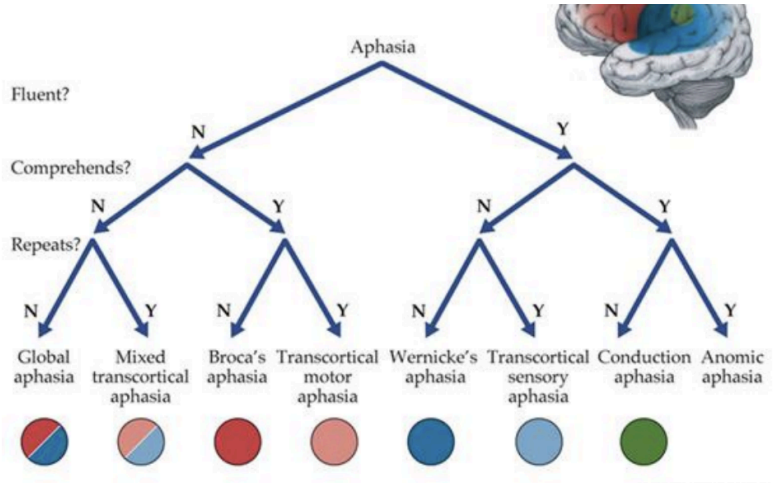
Study Object: Synchronic - at a single point in time - grammar/phonology/syntax/variation(diff people group) - expl sys Diachronic - change over time - old/morden ENG - expl origion **Displacement Prevarication**: fake Reflexivity Vervet Monkeys: referential meaning, Arb partly + Cul Tra partly ｜Honey Bees/Waggle Dance方向 + 距离: Arb partly, encoded info/symbolic mapping + displa + Cul Tra | Bird sounds 用于 mating / territory 有pattern 无具体meaning **Arbitrariness** 反: onomatopoeia（拟声词)+ iconic signs + looong(非绝对arb) **Cul Tra** 挺：Genie no input + oralist programme fail | Human L is open sys(Animail: closed), Combinatorial structure(word->sent, Ani: fixed signals), Abstract meaning(不依赖当前刺激, Ani: stimulus-bound) | Semiotics (Saussure): a **signifier** (form（声音 / 词 / 手势 ）represents a **signified (concept)** Because linguistic signs are arbitrary, they must be **conventional (社会约定).** Peirce: iconic (非arb) - form **resemble**s meaning - onomatopoeia（woof, meow）+ 图画 + 模仿动作 Indexical(非arb) - directly **link**ed to meaning (cause / association) - 🚬 烟 → 火 + 指路 + 部分特征推断整体 Symbolic - arbitrary + conventional - words + **traffic lights** - L sys | **Discreteness** 挺：phoneme system + sign parameters | **Produ** 挺：recursion + novel sent creation + nativism（UG）+ rule-based system | sense without referent: unicorn | reference without sense/m: names | One Sense, Many Referents: dog, student, **Vague words**（模糊词）:with underspecified boundaries: tall, rich, old, **Deictic words**（指示词）:whose reference depends on context: Person deixis: I, you, we + Spatial deixis: here, there + Temporal deixis: now,yesterday,tom | A word with multiple senses and multiple referents: **Polysemy**（多义词 ）multiple related meanings: fork, mouth, ceiling + **Homonymy**（同形异义 ）Two words that have unrelated meanings but identical form: bark (树皮）Bachelor: unmarried man, degree, feudal knight, fur seal - synchronic（共时 ）看现代 speaker 感觉❌ 不相关 → homonymy, diachronic（历时 ）看历史发展✔ 有联系 → polysemy | **Denotation**（指称意义）✔ 如果否认是矛盾 / 不自然（#）cookie :literal / dictionary meaning of a word | **Connotation**（内涵意义 ）如果可以否认（不矛盾）:cultural or emotional associations of a word: 单身bachelor positive, spinster negative **Metaphor**（隐喻 ）similarity “He is a lion” + **Metonymy**（转喻 ）association “The White House” + Synecdoche（部分代整体）“hands” = workers + Extension（意义扩展）| **Hyponymy**（上下义关系）Hyponym（下义词 ） = 更具体 Superordinate（上义词 ） = 更一般 cottage → house, monarch->co-hyponyms: king, queen | **Synonymy**（同义关系）Register（语体差异）:formal/informal, Connotation（内涵差异）情感 / 评价不同, Collocation（搭配差异) Words that naturally occur together: big mistake not large | **Antonymy**（反义关系）:Binary / Complementary antonyms: one **negates** the other + Gradable antonyms: Opposites on a **continuous scale** + Relational opposites: same relationship from opposite perspectives | **Taxonomic sisters** are co-hyponyms under the same superordinate成员不是关系: days of the week, diff genders, ethnicity | **Auto-antonyms**（自反义词 **/ Contronyms**）:a single word that has two opposite meanings, content-dependent: dress, clip固定, dust, seed ｜ **Wug test**: 挺 produ + rule-based sys + mental grammar | **morphemes**, stru of words: morphology, Taxonomy(分类 ）：Free morphemes - Can stand alone as a word : as root / base, carry core m + Bound morphemes: (功能）**Inflectional** (改变语法）-s, -ed + **Derivational**（改变词义词性）un-, -ness | Affixes(位置）:Prefix（前缀 ）Suffix（后缀 ）Infix（中缀a-**bleep**-mazing ｜Combining: **Acronyms**: NASA / **Abbreviations**: ASAP | **Blending:** smog, brunch | **Affixation**（加词缀 ）| **Compounding**: blackboard, smartphone | Remving: **Clipping/abbreviation/shortening**: ad, exam |**Backformation**: televise, edit 改变词类｜ grab: **Zero derivation/Conversion**: google n->v | **Borrowing**: sushi, cafe | **Eponyms**（专名词 ）sandwich（人名）, boycott | **Retronyms** arise from semantic expansion: phone->mobile phone->landline phone Semantic Meaning Change：**Broadening**:从具体 → 抽象 / 更广 companion, frock, sick: **Broadening via metaphor** | **narrowing**: 意义变更具体 meat | **Shift**: spinster | Affective meaning change: **Elevation**（more positive) awfully,terribly | **Deterioration** (more negative) cunning, spinster | **Lexicon = open sys**, 一个 lexical entry 包括：form + m + pronou + gram info | Feral children: Victor在法国森林中被发现（约12岁 ）几乎没有语言输入只理解简单词错过了 **critical period, Isabelle**和聋哑母亲隔离成长~6岁被发现完全恢复语言能力关键期内 → 可完全习得

Genie’s case suggests that language is not completely culturally transmitted, as she was unable to fully acquire grammar despite intensive exposure later in life. This supports the critical period hypothesis, indicating that there is a limited window for language acquisition. However, her ability to acquire some vocabulary suggests the presence of innate learning mechanisms. A caveat is that Genie’s prior developmental condition is unclear, which limits the strength of the conclusions. | **Ape** train speech: **Viki** - 生理结构限制（vocal tract）| train ASL: Washoe / Nim / Koko / Michael - 学会一些 signs, 缺乏语法（grammar）没有hierarchy, 多为 imitation（模仿）,but a little creativity| train lexigrams - Sarah / Lana / Kanzi - 理解能力较强（特别是 Kanzi）仍然没有复杂 syntax | highly train related, reward related | **Genie** behaves like human children in her ability to acquire vocabulary and use language communicatively, but resembles apes in her lack of grammatical structure and limited productivity, suggesting that both innate capacity and early exposure are necessary for full language development.

Co-speech gestures: alongside speech to support meaning: Iconics 手往上移动 rocket went up | Metaphorics 手往上 Prices went up | Deictics 指向位置 Put the box there | Beats 手势强调节奏 You! Are! Not! Listening! | **Symbolic gestures** = stand alone and convey meaning without speech: Pantomimes 模仿骑马,可以单独使用 ｜ Emblems👍 → "OK / good" 支持 arbitrariness 支持 cultural transmission ｜ Oralist programme：residual hearing，lipreading，speech production ❌ sign language ｜ 没有足够语言输入信息极不完整，language = grammar + structure + meaning发音训练 ≠ 语言习得，输入不足 →❌ 无法在关键期内习得语言，cognitive impact没有语言影响思维发展 | Not really Sign L: **SEE** 模拟英语结构 必须逐词表达 | **Fingerspelling**用手势表示字母, only 专有名词,新词,不是完整语言系统 ｜**Sign L**: hand shape手指变化19 ｜ location/place of articulation 手在脸的绝对位置12 | movement 移动方向24 ｜Orientation手掌朝向6 ｜ **NSL**: older pidgin no gram, younger pidgin → creole has gram 挺critical period | Innate sytax building + Regularisation: impose consistent structure on inconsistent input | language acquisition is largely implicit: **Imitation** 反 Children produce novel sentences，**Overgeneralisation** children apply rule｜ **Reinforcement** reward and punishment 反 Parents rarely correct grammar, mainly m, Language learning 太快输入不足输出复杂 | **Innate hypothesis** biologically equipped for language 挺 Wug, NSL, Overgeneralization | **Observational Method: CHILDES** Database: transcripts, age, children L data 语言发展阶段 大规模 corpus 跨语言 / 跨研究 不干预 记录自然语言使用 | experiment 控制变量 | 胎儿阶段（**Prenatal stage**）区分母亲声音记住熟悉语音 | 出生后（**Prelinguistic stage**）Cooing ~2–3个月 元音-like 声音 + Babbling ~6个月 类似语言的音节bababa | Receptive vocabulary (can understand)≈ 2 × productive vocabulary (can use) | **Overextension**通常理解: a word to refer to a broader set of objects | **Underextension**可能不完全理解: a word to refer to a narrower set of objects 支持 prototype theory categories不严格, 支持 active learning actively forming categories, 反驳 imitation theory创造用法 不是模仿 ｜ **Holophrastic Stage** 12–18 个月 one word full meaning 语法能力未成熟，认知 > 表达，依赖语境 ｜ **two-word stage**18–24个月两词组合 + 初步语法｜**telegraphic stage** 24–36个月 多词句 + 缺功能词 ｜3–4 years: acquiring inflectional morphemes and the marking of regular plurals ("one elephant begins to dance; two elephants begin to dance") ｜**dyslexia** phonological processing 问题 visual processing 问题 ｜ **CDS** for language development: phonology: higher pitch, exaggerated intonation + lexicon: simple, repetition + syntax: short, simple + interaction: more Qs, turn-taking,feedback | **Pet DS** is emotional: no gram, no learning aim, low interaction | **Elderly DS** for comprehension support: normal intonation **Nativism** Chomsky: **LAD** 不需要 formal teaching 需要input（exposure）only works in Critical Period |**UG**: categories(n, v, adj), structural rules children利用 UG+ 输入→ 推断具体语言规则 | Statistical Learning: children 不需要 UG 通过输入发现规律 - Pattern detection: 发现哪些词一起出现, Probability tracking 统计哪些组合常见, Chunking 把连续输入分成单位, Critical period is due to **general cognitive changes**, not language-specific mechanisms: brain flexibility decreases, memory / attention changes, first language interference | Psycholinguistic: **Stroop task**->reading is automatic and difficult to suppress: congruent red RED, incongruent blue RED, unrelated red THIN, non-word red KUD, 测RT和Accuracy ｜ **music experienment**: 测**brainstem activity**，confounding variables: Innate musical ability, SES, "Mi" is a musical note | McGurk effect: Visual + Auditory vs Auditory only-> When visual and auditory signals conflict, perception is altered**, participants self-report | Phonemic Restoration Effect** 实际没有声音但是听到了：缺失音 + 静音 很容易发现缺失 vs 缺失音 + 噪音 听起来是完整的 -> Top-down processing大脑用知识来猜, Context effect句子越合理越容易预测, automatic自动补全 ｜**slips of the tongue** -> Forward planning, levels of processing, open-class words(n,v) | **LENA system**: recording caregiver speech quantity, child vocalisations,turn-taking quantitative data -> Input amount matters | **Governmental statistics** 教育水平 家庭背景 socioeconomic status | **online = processing**过程, offline = 结果/判断 | **Eye-tracking** 在 open-class words 上停留更久, grammatical words 处理更快, Dyslexia fixation 更长 回看更多（regression）reading slower | **PET** blood flow, fMRI BOLD signal, **EEG** eletronic act N400: semantically or pragmatically inappropriate words, P600: grammatically inappropriate words, baby｜**Gaze / visual world paradigm -> attention | LH** 语言结构,语法 **/** 词汇 **RH** 情绪语调（emotional prosody）音乐 整体感知 contralateral connection **Corpus callosum** | **prosody** has both linguistic and non-linguistic aspects, and is processed by both hemispheres, though often right-hemisphere dominant. 语调（intonation）重音（stress）节奏（rhythm）右视野 左脑 能说 能画 先说，左视野 右脑 不能说 左手能画 只能画 ｜**Broca** left frontal gyrus BA44 syntax BA45 semantic processing - speech production,syntax - non-fluent, relatively preserved comprehension, telegraphic speech(no gram) | **Wernicke** posterior superior temporal gyrus BA22 (superior temporal sulcus,angular gyrus（BA39）supramarginal gyrus（BA40）) - comprehension meaning - fluent 理解困难 nonsense speech | **Arcuate fasciculus**链接两个区域 -> **Conduction aphasia** 理解说话正常不能重复 ｜ **Neuroplasticity**: organise its structure and function based on experience - 技能学习,Echolocation（回声定位 | **Compensation** refers to the brain's ability to reorganise and allow undamaged areas to take over functions of damaged regions. Neuroplasticity 是基础 需要经验

physiotherapy,rehabilitation不是自动发生 - Hemispherectomy（半球切除）年龄越小 → 代偿能力越强| **lesion studies** 不完全可靠 Brain damage does not necessarily reflect the original functional organisation of the brain 解决方法：**TMS** “Virtual lesion”（虚拟损伤 ）突然发生（crucial!!）可定位（localized - 无 compensation 干扰 更接近因果关系 可重复 | **Priming**: Faster processing of a target word due to prior exposure to a related stimulus (prime) - Semantic Priming意义相关，Phonological priming sound相似，Morphological priming 词根相关，Repetition priming同词重复 - mental lexicon 是网络spreading activation model，支持 online processing ｜**Lexical retrieval** is affected by: ⁕ meaning ⁕ sound ⁕ frequency⁕ How concrete the word is ⁕ Context ｜**Spreading Activation Model**：跨语言共享网络，translation links 最强 同一个概念，concrete 更强有感知基础（imageable）更容易激活 ｜**Revised Hierarchical Model**（**RHM**）：low proficiency - L2 → L1 → concept - L2 meaning is accessed via L1，更依赖 form，high proficiency - L2 → concept（直接）- L2 is directly linked to meaning，更依赖 meaning ｜**Dong 2005** - 用英语时（L2）bilingual更像 English monolingual proficiency 越高 → 越像，用中文时（L1）bilingual：几乎和 Chinese monolingual 一样 - 多语言部分共享 + 部分差异，支持 linguistic relativity，bilingual cognition 是动态的 ｜Concreteness may affect how easily a L2 word is linked to the L1 conceptual network. ⁕ **Conceptual network** may also become differentiated into overlapping components and components that are specific to each language, e.g. language-specific connotations.



Semantic Components：binary（常用 + / −）Minimal pairs differ in only one semantic feature ｜**componential analysis**：4 - 2, 5-8 - 3, 9-16 - 4 加1个all yes cate | **Prototype Theory**: Category membership is based on degree of similarity to a prototype - 1 中心 vs 边缘 central members（典型）peripheral members（边缘 ）2 category boundaries 模糊 3 判断速度不同 典型 → 判断更快 - Word frequency 更常见但评分低）Personal preference 参与者偏好不同但评分一致 Appearance 高评分项目外观差异很大 外观相似却评分不同 | By binary semantic components • Componential analysis is a way of decomposing words into their semantic components. • Useful in articulating the elements of a semantic category and how words are related in terms of meaning • Some issues include: - How semantic components are labelled. - Words can be used loosely or informally. - People’s conceptual categories may not involve a checklist of binary features. - Categories may have fuzzy boundaries. - Some categories are not defined by a clear set of criteria, but by family resemblance. | By prototypes • Prototype theory proposes that category membership is determined by how similar items are to a prototype. - Untypical exemplars may still share certain central features with the prototype, which may be related to appearance, function, etc. • Issues: - Not clear how prototypes are identified or determined. - Some conceptual categories do not involve prototypes. ➢ Componential analysis and prototype theory complement each other – categorisation may involve multiple components.

Sapir–Whorf Hypothesis: Strong version（linguistic determinism 反 人可以理解新概念，翻译是可能的，非语言认知存在（如视觉 ）**Weak version**（linguistic relativity）挺：colour terms，空间（Boroditsky），时间表达, | Choi 语言影响随发展增强初期 universal后期 language-shaped | Attention → Encoding → Memory | linguistic metaphors = 表面, conceptual metaphor = 深层结构, Mapping（映射）把一个领域（source）映射到另一个领域（target）具体到抽象